Mo Tu We Th Fr Sa Su

AI System Design

Date 12/10/26.

1) Core Building Blocks of Production LLM.

(i) User Interface:

+ It is the primary interface where the user will interact with my application.

UI → backend → Server/API

2) Why can't frontend alone direct access the API/server?

+ Security Reasons:
  - While frontend uses an API key to access my service provider, it is highly exposed & could easily be stolen.
  - Data (public/private) could be exposed

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Even if frontend uses Https, the risks don't disappear.

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(ii) API Layer:

→ API layer manages the incoming requests.

Ex) If the client UI sends:

POST /chat

{ "question": ---

"doc_id": --- }

Now, a series of things could be done by this layer:

* Authentication → who makes the req?
* Authorization → Does user have access?
* Rate limit → Has the user exceeded req limit?
* Request validate → Is the req. body valid?
* Session Manage → Which convo this req. belong to?
* Http Response → How must backend respond?

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(iii) Input Guardrails:

→ This layer is responsible for checking the input & identifying the prompts given by the user is actually safe or not.

→ The App must have a set of Rules & Regulations, which must check whether the i/p is safe/not.

ex) User asks, "How to make a Bomb" - the System must not process it. Also, if user asks to forget all the rules set by the creator the system must identify that too.

Likewise, it ensures i/p safety by setting up different things such as:

* Input Validation (Malformed/Large i/p detect)
* System instructions (Guidelines to follow a certain behaviour)
* Detect suspicious instructions/Prohibited requests

Whatever your goal, you can get there if you're willing to work